

Ghana Communication Technology University

Course Outline



Faculty of Computing and Information Systems

CSSD 204: SCRIPTING LANGUAGES

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COURSE DESCRIPTION

This course introduces students to scripting languages with a focus on automating tasks, rapid application development, and integration with other software systems. It emphasizes practical skills using popular scripting languages such as **Python, JavaScript, and Bash**. Students will gain experience in writing scripts for web development, data manipulation, system administration, and software prototyping.

COURSE OBJECTIVES AND LEARNING GOALS

Key Topics:

- To provide students with an understanding of the nature and role of scripting languages.
- To introduce students to some popular scripting languages and their applications.
- To enable students to write simple scripts using these languages for a variety of applications.

Learning Outcomes:

By the end of this course, students will be able to:

1. Differentiate between compiled and interpreted languages
2. Write scripts using Python, JavaScript, and Bash
3. Use scripting to automate system tasks and data processing
4. Develop basic web applications using scripting languages
5. Interface scripts with other software and APIs

RESOURCES

- Supplementary readings and resources will be provided throughout the course.

WEEK-BY-WEEK TOPICS

Weekly Activities

Each week you will need to:

- Attend the lectures
- Read recommended pages in the textbook (listed below)
- Complete the interim assessment for feedback
- Complete the homework assignments
- Participate in discussions

SCHEDULE:

Week	Module	Topic	Readings
1	1	Introduction to Scripting languages: Overview of Scripting vs. programming languages Use cases of Scripting (web, automation, system administration)	
2	2	Overview of Python: History and features Installing Python and setting up the environment Python Basics: Variables, data types, and operators Control structures (if, loops)	

3	3	Advanced Python Concepts: Functions and Modules - Defining and calling functions - Importing modules and libraries File Handling - Reading and writing files (I/O) - Exception handling (try/except)	
4	4	Python for Automation: Scripting repetitive tasks Working with libraries (e.g., os, shutil, datetime) Automating file and directory operations	
5	5	Introduction to Bash Scripting: Shell basics and Linux commands Writing and running bash scripts Variables, conditionals, loops in bash	
6	6	Bash Scripting for System Tasks Automating backups and logs Scheduling tasks with cron Using pipes and redirection	
7	7	Introduction to JavaScript JS syntax and variables Functions and scope	
8	8	JavaScript for Web Development DOM manipulation basics Events and event handling Form validation Fetch API and async operations	
9	9	Scripting with APIs and JSON RESTful APIs and data formats Consuming APIs with Python (e.g., requests)	

		Consuming APIs with JavaScript (fetch/axios)	
10	10	Regular Expressions in Scripting Pattern matching in Python, JS, and Bash Real-world examples: validating emails, parsing logs	
11	11	Interfacing Scripts with Databases Introduction to SQLite or MySQL Database access using Python CRUD operations via script	

GRADING CRITERIA

Assessment type	Grade
Assignments / Homework	30%
Class Attendance (virtual inclusive)/ Participation	10%
End-of-semester exams	60%

GRADING SYSTEM

Score	Grade	Interpretation
100 – 80	A	Outstanding
79 – 75	A -	Excellent
74 – 70	B +	Very Good
69 – 65	B	Good
64 – 60	B -	Fairly Good
59 – 55	C +	Above Average
54 – 50	C	Average
49 – 45	C -	Below Average
44 – 40	D	Pass
39 – 0	F	Fail
	X	Absent
	Z	Disqualified
	IC	Incomplete
	AD	Audit